

Stackup: JLC04161H-7628

	Material	Layer	Thickness	Dielectric	Type	Gerber
		F.Paste			Paste Mask	
		F.Silkscreen			Legend	GBR
		F.Mask	0.01mm		Solder Mask	GBR
	Copper	L1 (Sig)	0.035mm (1oz)		Signal	GBR
	Prepreg		0.21mm	7628	Dielectric	
	Copper	L2 (GND)	0.015mm (0,42857142857142855oz)		Plane	GBR
	Core		1.06mm	FR4	Dielectric	
	Copper	L3 (PWR)	0.015mm (0,42857142857142855oz)		Plane	GBR
	Prepreg		0.21mm	7628	Dielectric	
	Copper	L4 (Sig.)	0.035mm (1oz)		Signal	GBR
		B.Mask	0.01mm		Solder Mask	GBR
		B.Silkscreen			Legend	GBR
		B.Paste			Paste Mask	

Total thickness: 1.6mm
Note: external layer thicknesses are specified after plating

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
USB	90	±10 %	L1	0.17	0.275	L2

- | | |
|------------------|---------------------|
| BOARD SIZE | 90.000 x 100.000 mm |
| BOARD THICKNESS | 1.600 mm |
| TRACE WIDTH | 0.170 mm |
| TRACE TO TRACE | -0.000 mm |
| MIN. HOLE (PTH) | 0.300 mm |
| MIN. HOLE (NPTH) | 0.650 mm |
| ANNUAL RING | 0.075 mm |
| COPPER TO HOLE | 0.150 mm |
| COPPER TO EDGE | 0.300 mm |
| HOLE TO HOLE | 0.250 mm |

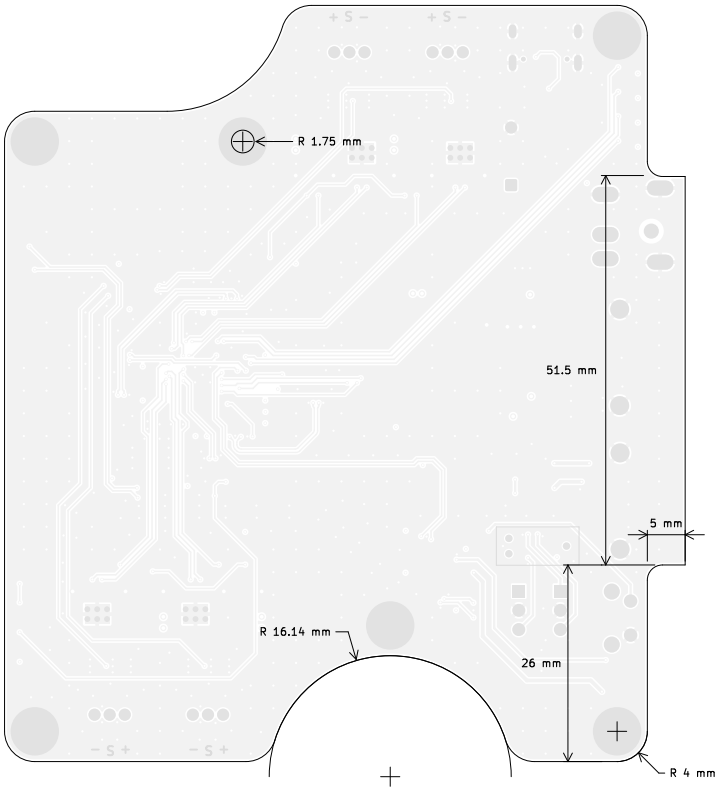
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

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KiCad E.D.A. 9.0.1+1 + KiBot v1.8.5

Bottom Fabrication (Scale 1:1)



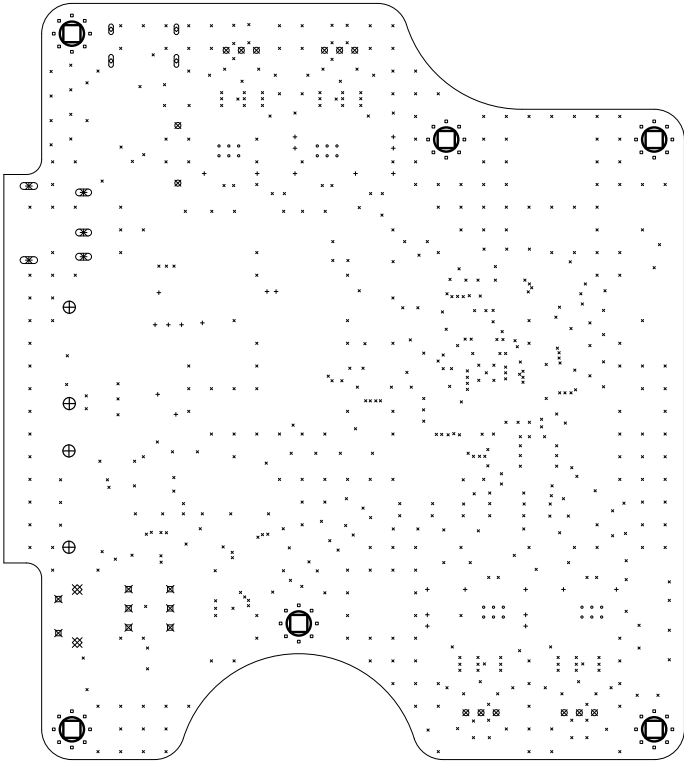
All dimensions are in millimeters unless otherwise specified.

Designed for: TrendBit Designed by: Petr_Malaník					
Sheet: File: pump_board.kicad_pcb				Project: SMPBR	
Title:				Board: Pump_board	
Size: A4		Date: 2024-12-04		GIT hash: e67dce0	
KiCad E.D.A. 9.0.1+1 + KiBot v1.8.5				Rev: 1.0	
4		5		Id: 2/10	

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	648	0,30mm (11,81mils)	PTH	Round	L1 (Sig.) - L4 (Sig.)	Via
○	24	0,30mm (11,81mils)	PTH	Round	L1 (Sig.) - L4 (Sig.)	Pad
+	27	0,50mm (19,69mils)	PTH	Round	L1 (Sig.) - L4 (Sig.)	Via
□	48	0,50mm (19,69mils)	PTH	Round	L1 (Sig.) - L4 (Sig.)	Pad
◇	4	0,60mm (23,62mils)	PTH	Slot	L1 (Sig.) - L4 (Sig.)	Pad
⊗	14	0,80mm (31,50mils)	PTH	Round	L1 (Sig.) - L4 (Sig.)	Pad
✱	5	0,90mm (35,43mils)	PTH	Slot	L1 (Sig.) - L4 (Sig.)	Pad
⊠	8	1,00mm (39,37mils)	PTH	Round	L1 (Sig.) - L4 (Sig.)	Pad
⊗	2	1,30mm (51,18mils)	PTH	Round	L1 (Sig.) - L4 (Sig.)	Pad
⊕	4	1,60mm (62,99mils)	PTH	Round	L1 (Sig.) - L4 (Sig.)	Pad
○	6	3,20mm (125,98mils)	PTH	Round	L1 (Sig.) - L4 (Sig.)	Pad
Total 790						



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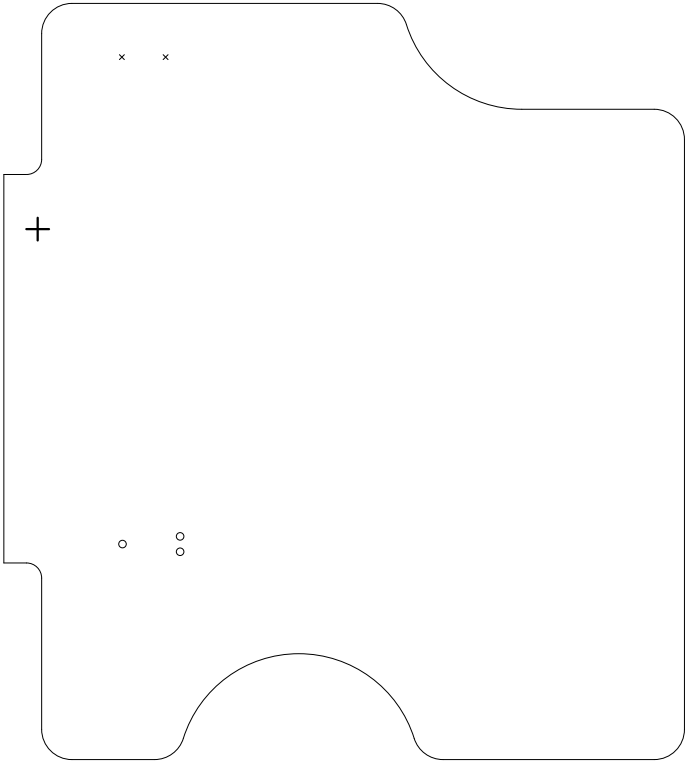
Project: SMPBR
Board: Pump_board

Title:
Size: A4 Date: 2024-12-04 GIT hash: e67dce0 Rev: 1.0
KiCad E.D.A. 9.0.1+1 + KiBot v1.8.5 Id: 3/10

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	0,65mm (25,59mils)	NPTH	Round	L1 (Sig.) - L4 (Sig.)	Mechanical
○	3	1,00mm (39,37mils)	NPTH	Round	L1 (Sig.) - L4 (Sig.)	Mechanical
+	1	3,00mm (118,11mils)	NPTH	Round	L1 (Sig.) - L4 (Sig.)	Mechanical
Total 6						



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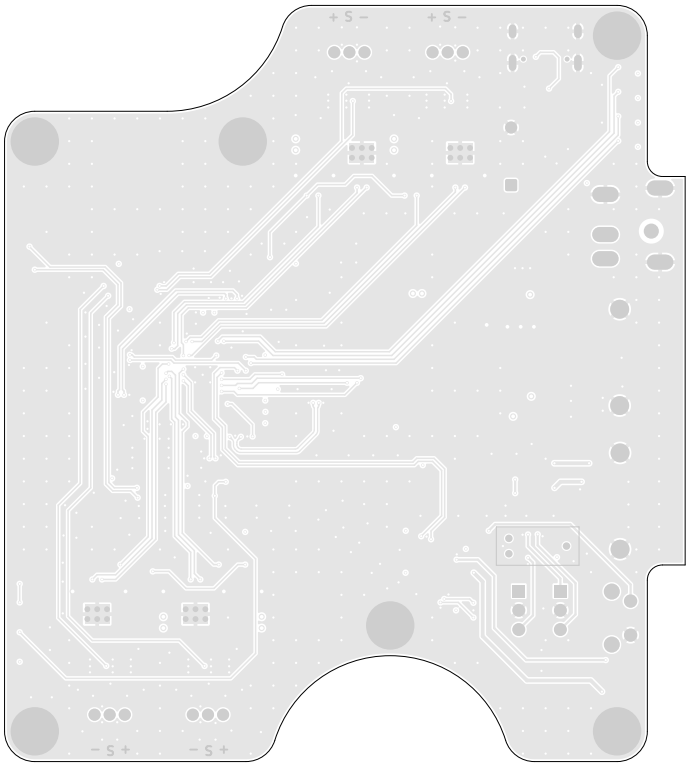
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Board: Pump_board

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Size: A4 Date: 2024-12-04 GIT hash: e67dce0 Rev: 1.0
KiCad E.D.A. 9.0.1+1 + KiBot v1.8.5 Id: 4/10

			
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KiCad E.D.A. 9.0.1+1 + KiBot v1.8.5			Id: 5/10

Bottom Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
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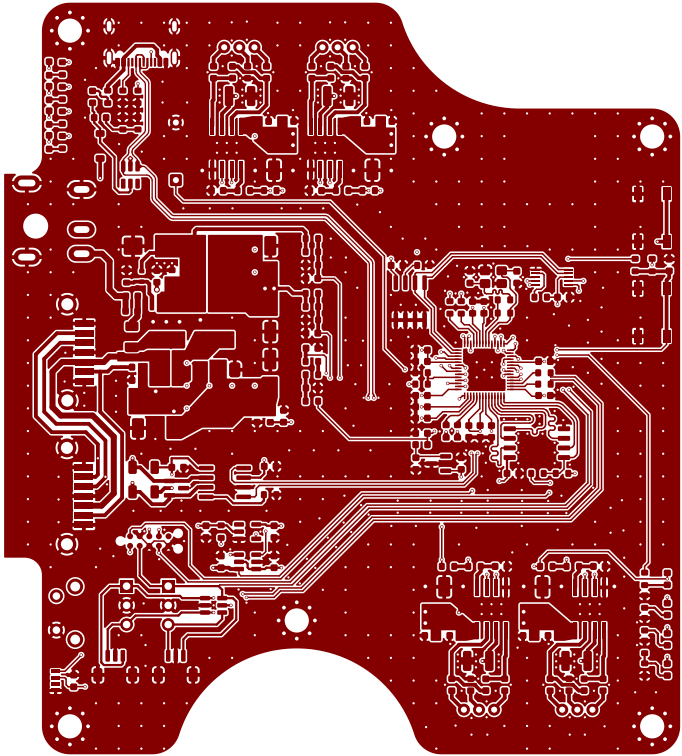


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File: pump_board.kicad_pcb

Project: SMPBR
Board: Pump_board

Title:		Date: 2024-12-04		GIT hash: e67dce0		Rev: 1.0
Size: A4		KiCad E.D.A. 9.0.1+1 + KiBot v1.8.5		Id: 6/10		

L1 (Sig) (Scale 1:1)

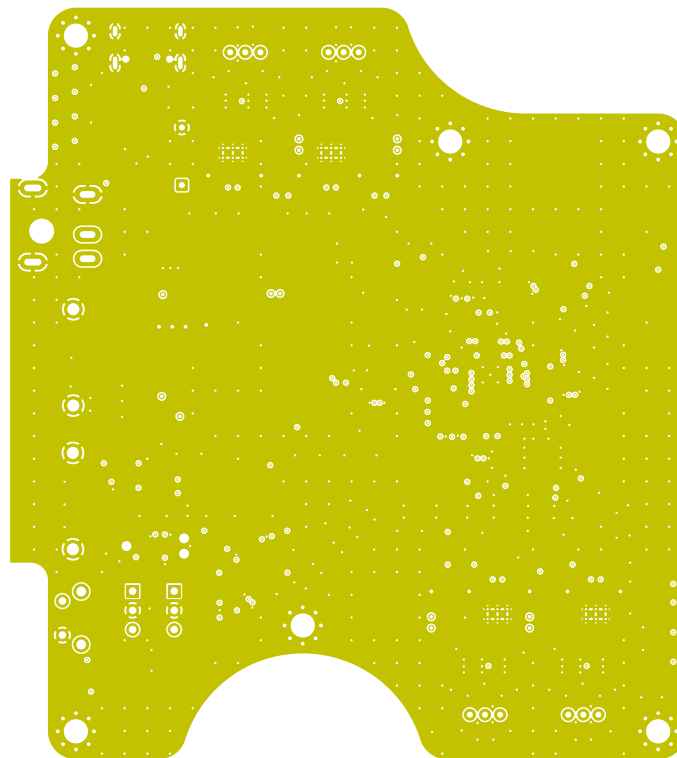


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			Board: Pump_board	
Size: A4	Date: 2024-12-04	GIT hash: e67dce0	Rev: 1.0	
KiCad E.D.A. 9.0.1+1 + KiBot v1.8.5			Id: 7/10	

L2 (GND) (Scale 1:1)



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Title:

Project: SMPBR

Board: Pump_board

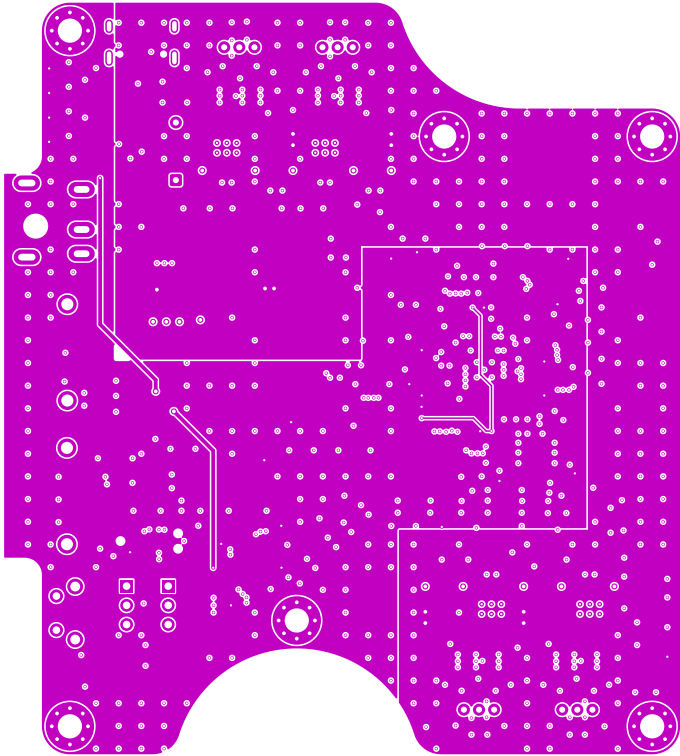
Size: A4
KiCad E.D.A. 9.0.1+1 + KiBot v1.8.5

Date: 2024-12-04
GIT hash: e67dce0

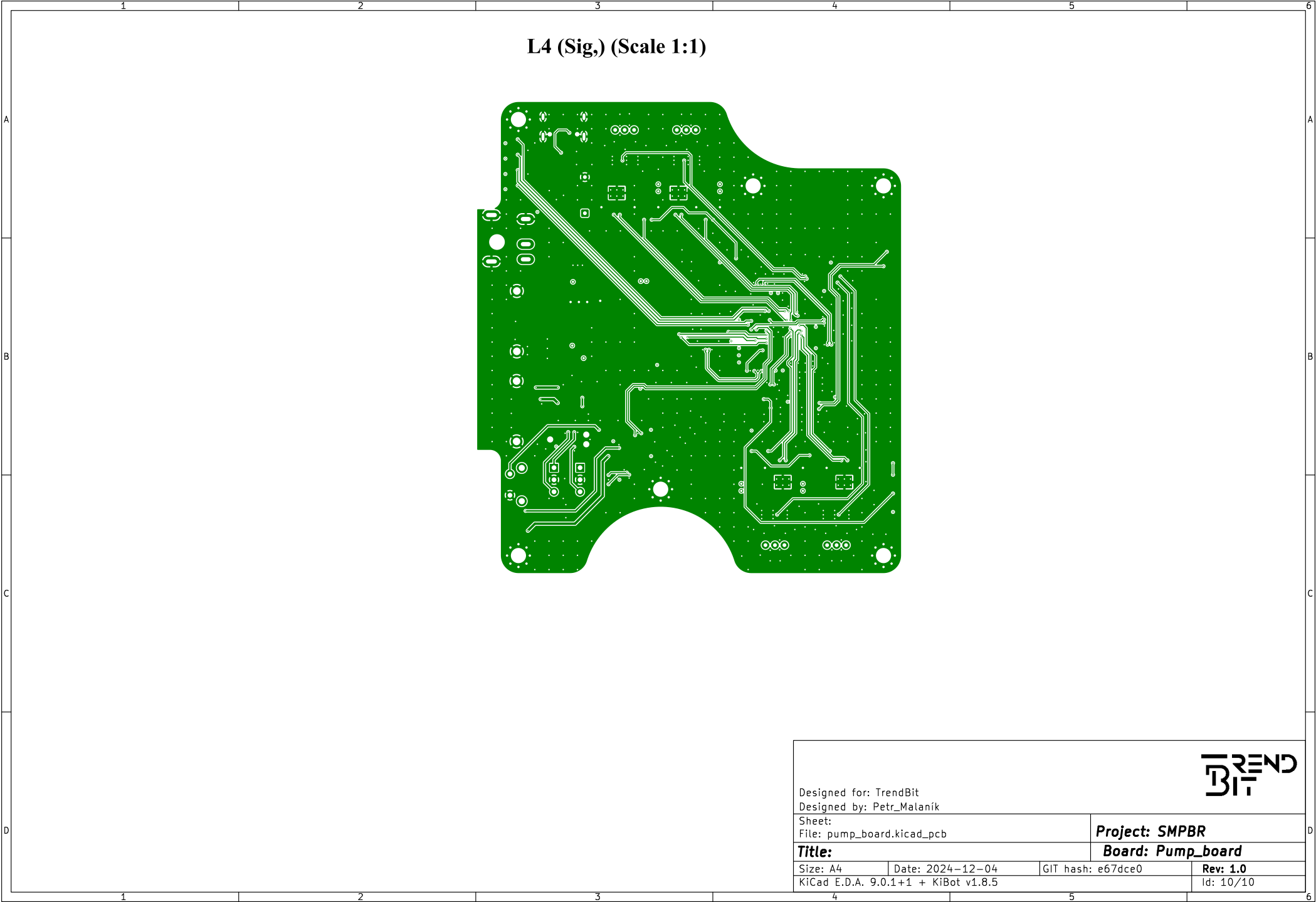
Rev: 1.0
Id: 8/10

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L3 (PWR) (Scale 1:1)



				
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Size: A4	Date: 2024-12-04	GIT hash: e67dce0		Rev: 1.0
KiCad E.D.A. 9.0.1+1 + KiBot v1.8.5				Id: 9/10



				
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Size: A4	Date: 2024-12-04	GIT hash: e67dce0		Rev: 1.0
KiCad E.D.A. 9.0.1+1 + KiBot v1.8.5				Id: 10/10